Power converters



文件: power.kicad_sch

Panel and temperature



文件: panel.kicad_sch

Host interface and test points



文件: interface.kicad_sch

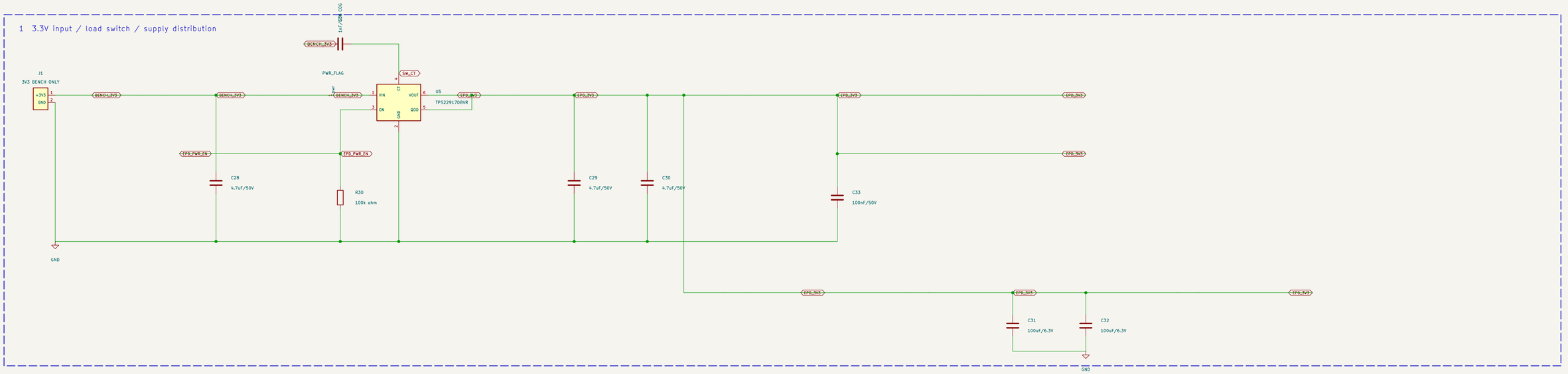
album

Sheet: /
File: gdep133c02-driver.kicad_sch

**Title: GDEP133C02 minimum driver - DRAFT**

| Size: A4 | Date: 2026-10-09 | **Rev: 0.10-E01** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/4 |

## 1 3.3V input / load switch / supply distribution




v0.4: U5 output = EPD_3V3; direct copper feeds to logic and power branches

## 2 Positive source rail (VDDP)



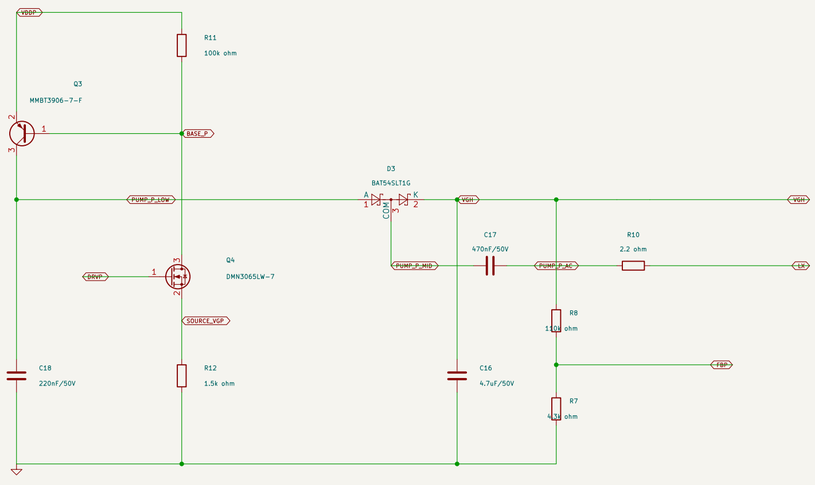

## 3 Negative source rail (VDDN)



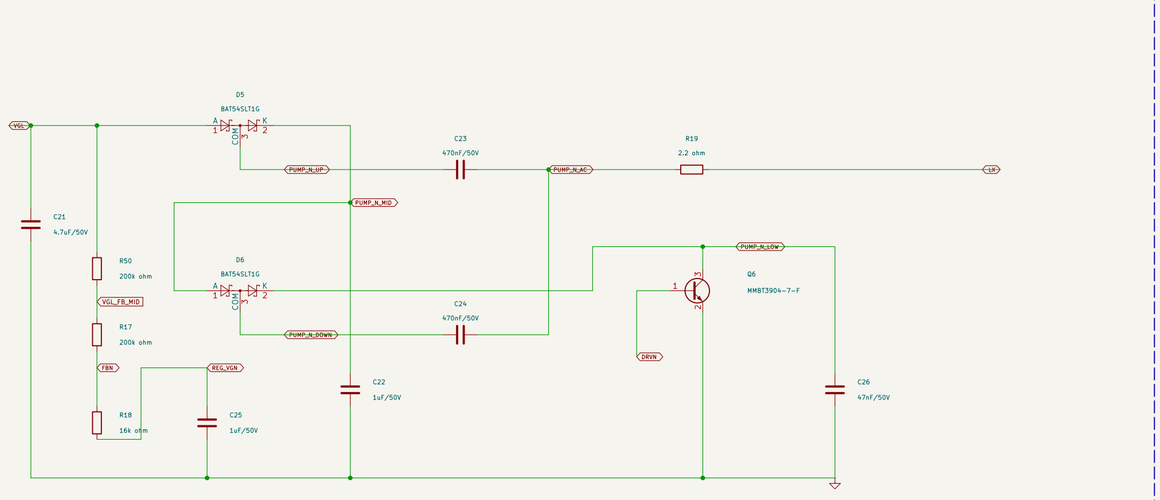

## 4 VCOM supply branch




## 5 Positive gate charge pump (VGH)




## 6 Negative gate charge pump (VGL)




Dots join wires; crossings without dots are not connected. Net labels retain the existing PCB net names.

| | |
|---|---|
| Sheet: /Power converters/ | |
| File: power.kicad_sch | |
| **Title: Power – wired functional circuits** | |
| Size: User | Date: 2026-10-09 |
| KiCad E.D.A. 10.0.6 | Id: 2/4 |
| | **Rev: 0.10-E01** |

## 1 Screen connector — pin numbers retained; functional pin layout

FPC1

FPC-05FB-60PH20

POWER

LOGIC / SPI

GATE / SOURCE RAILS

GATE / SENSE

SOURCE RAILS

| Pin | Net | Pin | Net |
|---|---|---|---|
| 9 | EPD_3V3 | 20 | TSCL |
| 10 | EPD_3V3 | 21 | TSDA |
| 19 | EPD_3V3 | 22 | BS0 |
| 12 | VCC | 23 | BS1 |
| 18 | VCC1 | 24 | RES_N |
| 13 | VDDP | 25 | BUSY_N |
| 14 | VDDP | 27 | CS_M_N |
| 16 | VDDN | 28 | SCLK |
| 17 | VDDN | 29 | SI0 |
| 59 | VGH | 30 | SI1 |
| 60 | VGL | 31 | SI2 |
| 41 | VNCP_3P5V | 32 | SI3 |
| 56 | VBB_3P5V | 58 | CS_S_N |
| 1 | VCOMBD_M | 54 | FPL_VCOM |
| 36 | VCOMBD_S | 55 | TFT_VCOM |
| 2 | RESEC | 42 | VSPH |
| 3 | GDRC | 45 | VSPH |
| 4 | RESEN | 43 | VSPL |
| 5 | GDRN | 46 | VSPL |
| 6 | RESEP | 44 | VSPL2 |
| 7 | GDRP | 47 | VSPL2 |
| 33 | DRVP | 48 | VSNH |
| 34 | FBP | 53 | VSNH |
| 37 | DRVN | 49 | VSNL |
| 38 | FBN | 52 | VSNL |
| 39 | REG_VGN | 50 | VSNL2 |
| 8 | GND | 51 | VSNL2 |
| 26 | GND | 11 | NC |
| 35 | GND | 15 | NC |
| | | 40 | DRVN2 |
| | | 57 | NC |

C7 10uF/50V X5R

C10 470nF/50V

C11 470nF/50V

R20 4.7k ohm

R21 4.7k ohm

## 2 SPI mode / signal defaults

R1 2k ohm

R35 10k ohm

C9 470nF/50V

R36 100k ohm

C8 4.7uF/50V

## 3 Panel-controlled temperature sensor

U4

TCN75AVDA713

| Pin | Net |
|---|---|
| VDD | EPD_3V3 |
| SCL | TSCL |
| SDA | TSDA |
| ALERT_N | 3 |
| A2 | GND |
| A1 | GND |
| A0 | GND |
| GND | GND |

C27 100nF/50V

R23 DNP: alert output is not strapped to ground.

## 4 Source outputs / input straps / capacitor banks

| | | | |
|---|---|---|---|
| C1 10uF/50V X5R | C103 10uF/50V X5R | C104 10uF/50V X5R | C105 3.3uF/50V |
| C2 10uF/50V X5R | C106 10uF/50V X5R | C107 10uF/50V X5R | C108 3.3uF/50V |
| C3 10uF/50V X5R | C109 10uF/50V X5R | C110 10uF/50V X5R | C111 3.3uF/50V |
| C4 10uF/50V X5R | C112 10uF/50V X5R | C113 10uF/50V X5R | C114 3.3uF/50V |
| C5 10uF/50V X5R | C115 10uF/50V X5R | C116 10uF/50V X5R | C117 3.3uF/50V |
| C6 10uF/50V X5R | C118 10uF/50V X5R | C119 10uF/50V X5R | C120 3.3uF/50V |

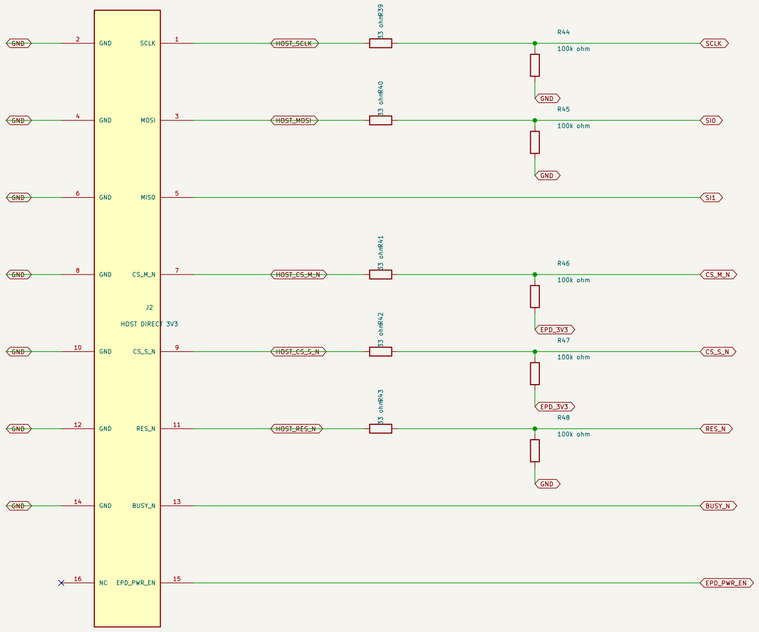
Nominal 33.3uF; effective capacitance pending.

Symbol grouping does not represent physical FPC pin positions.

Dots join wires; crossings without dots do not connect. Global labels are cross-sheet interfaces.

| | |
|---|---|
| Sheet: /Panel and temperature/ | |
| File: panel.kicad_sch | |
| **Title: Panel — wired functional circuits** | |
| Size: User | Date: 2026-10-09 |
| KiCad E.D.A. 10.0.6 | **Rev: 0.10-E01** |
| | Id: 3/4 |

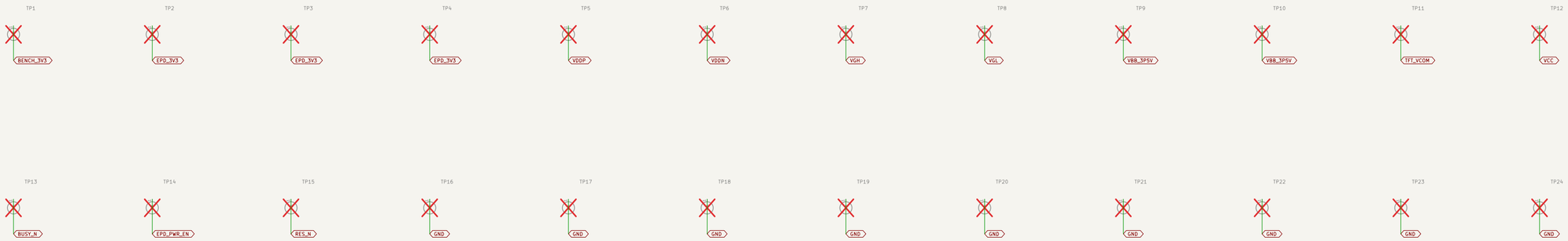
Direct 3.3V host interface — NO power-off signal isolation




Host first ON / last OFF. Screen OFF: no GPIO HIGH, no pull-ups.

J2.16 unused. No live plugging. Short cable <= 10cm.

Test pads: copper only, no parts to buy or solder




Sheet: /Host interface and test points/
File: interface.kicad_sch

**Title: Interface — wired functional circuits**

| Size: User | Date: 2026-10-09 | **Rev: 0.10-E01** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/4 |